

AUROADAS: A Modular Multi-Sensor Fusion and Perception System for ADAS

Emmanuel Musvaire

Department of Computer Science and Informatics

Oakland University

Rochester, MI, USA

Email: emusvaire@oakland.edu

Abstract

The increasing complexity of Advanced Driver Assistance Systems (ADAS) demands perception and localization pipelines that remain accurate, robust, and computationally efficient under diverse operating conditions. We present AUROADAS, a modular multi-sensor fusion and perception system built on ROS2 and a GTSAM factor-graph backend. AUROADAS integrates stereo vision, inertial measurements, GPS, wheel odometry, and optional LiDAR to produce stable real-time state estimates and object-level perception outputs. The system achieves less than 1% drift on KITTI and CARLA benchmarks, operates at 30–60 Hz with 22–28 ms latency on embedded hardware, and demonstrates resilience to sensor dropouts with recovery times under 0.5 s. An ablation study highlights the critical role of IMU and visual factors in maintaining accuracy, while mapping and loop closure significantly reduce long-term drift. These results indicate that AUROADAS provides a reliable and extensible foundation for production-level ADAS pipelines.

Index Terms

ADAS, Multi-Sensor Fusion, Factor Graphs, Visual-Inertial Odometry, Mapping, Perception, Autonomous Driving

1. Introduction

Motivation

Modern ADAS systems operate in environments characterized by dynamic motion, inconsistent lighting, partial occlusions, and intermittent sensor failures. These conditions expose fundamental weaknesses in traditional visual-inertial odometry pipelines, which rely heavily on continuous visual input and lack mechanisms for global correction. As vehicles increasingly depend on perception-driven decision-making, there is a critical need for architectures that combine redundancy, robustness, and real-time performance. AUROADAS is motivated by this

gap: it integrates multiple complementary sensor modalities within a factor-graph framework, enabling stable localization, resilient perception, and long-term consistency even when individual sensors degrade. By unifying visual front-end processing, multi-sensor fusion, mapping, and object-level perception, AUROADAS provides a practical and extensible foundation for next-generation ADAS systems.

Limitations of traditional VIO pipelines

Advanced Driver Assistance Systems (ADAS) rely on accurate and robust perception and localization to operate safely under diverse and often adverse real-world conditions. These conditions include sensor noise, dynamic occlusions, GPS degradation, illumination changes, and strict real-time constraints imposed by vehicle control loops [1], [2]. Traditional visual-inertial odometry (VIO) pipelines, while effective in controlled environments, tend to degrade significantly when subjected to long trajectories, intermittent sensor failures, or environments with limited visual texture [3], [4]. Their reliance on tightly coupled visual and inertial measurements makes them particularly vulnerable to camera dropouts, rapid motion, and inconsistent GPS availability.

Need for multi-sensor redundancy

To address these challenges, we introduce AUROADAS, a modular multi-sensor perception and localization system designed specifically for ADAS applications. AUROADAS integrates visual front-end processing, multi-sensor factor-graph fusion, mapping, and object-level perception into a unified architecture. The system leverages stereo vision, IMU preintegration, GPS constraints, wheel odometry, and optional LiDAR to maintain stable state estimation even under degraded sensing conditions. By employing incremental smoothing through a GTSAM-based factor graph [5], AUROADAS achieves both short-term motion stability and long-term global consistency. Furthermore, the mapping subsystem provides loop closure capabilities that mitigate drift accumulation over extended trajectories, while the perception module delivers obstacle detection, tracking, and confidence scoring suitable for ADAS decision-making [6].

Overview of AUROADAS

Overall, AUROADAS is designed to overcome the limitations of conventional VIO pipelines by embracing a multi-modal, redundancy-driven architecture that remains robust across diverse operational scenarios. The system's modularity, real-time performance, and resilience to sensor degradation position it as a practical foundation for production-grade ADAS perception and localization.

2. Related Work

ADAS Perception and localization

Perception and localization for ADAS and autonomous systems have been extensively studied across visual odometry, visual-inertial fusion, SLAM, and multi-sensor architectures. Classical visual odometry approaches such as ORB-SLAM and its variants have demonstrated strong performance in structured environments, but they degrade under low-texture scenes, dynamic occlusions, and rapid motion [1], [2]. Visual-inertial odometry (VIO) systems, including OKVIS and VINS-Fusion, improve robustness by tightly coupling camera and IMU measurements; however, they remain sensitive to camera dropouts, inconsistent lighting, and long-term drift due to the absence of global constraints [3], [4].

Factor-graph optimization

Factor-graph-based methods have emerged as a powerful alternative for multi-sensor fusion, offering modularity, incremental smoothing, and the ability to incorporate heterogeneous sensor modalities. GTSAM and iSAM2 provide efficient frameworks for integrating IMU pre-integration, GPS priors, wheel odometry, and visual reprojection factors in real time [5], [6]. These approaches have been applied in robotics, aerial vehicles, and autonomous driving, demonstrating improved consistency over extended trajectories.

Mapping and loop closure

Mapping and loop closure techniques further enhance long-term accuracy by correcting accumulated drift. Methods such as Scan Context, FAB-MAP, and pose-graph optimization have shown effectiveness in urban environments where repeated structures enable reliable place recognition [7], [8]. In parallel, ADAS perception research has advanced object detection, tracking, and scene understanding using both classical and learning-based approaches [9], [10].

How AUROADAS differs

AUROADAS builds upon these foundations by combining visual front-end processing, factor-graph fusion, loop-closure-based mapping, and object-level perception into a unified, real-time ADAS pipeline. Unlike traditional VIO systems, AUROADAS incorporates redundant sensor modalities and confidence-aware fusion, enabling robust operation under degraded sensing conditions.

3. System Overview

AUROADAS is designed as a modular, real-time perception and localization system optimized for ADAS requirements. The architecture consists of four primary subsystems: (1) Sensor Interface Layer, (2) Visual Front-End, (3) Multi-Sensor Fusion Layer, and (4) Mapping and Perception Layer.

3.1 AUROADAS Architecture

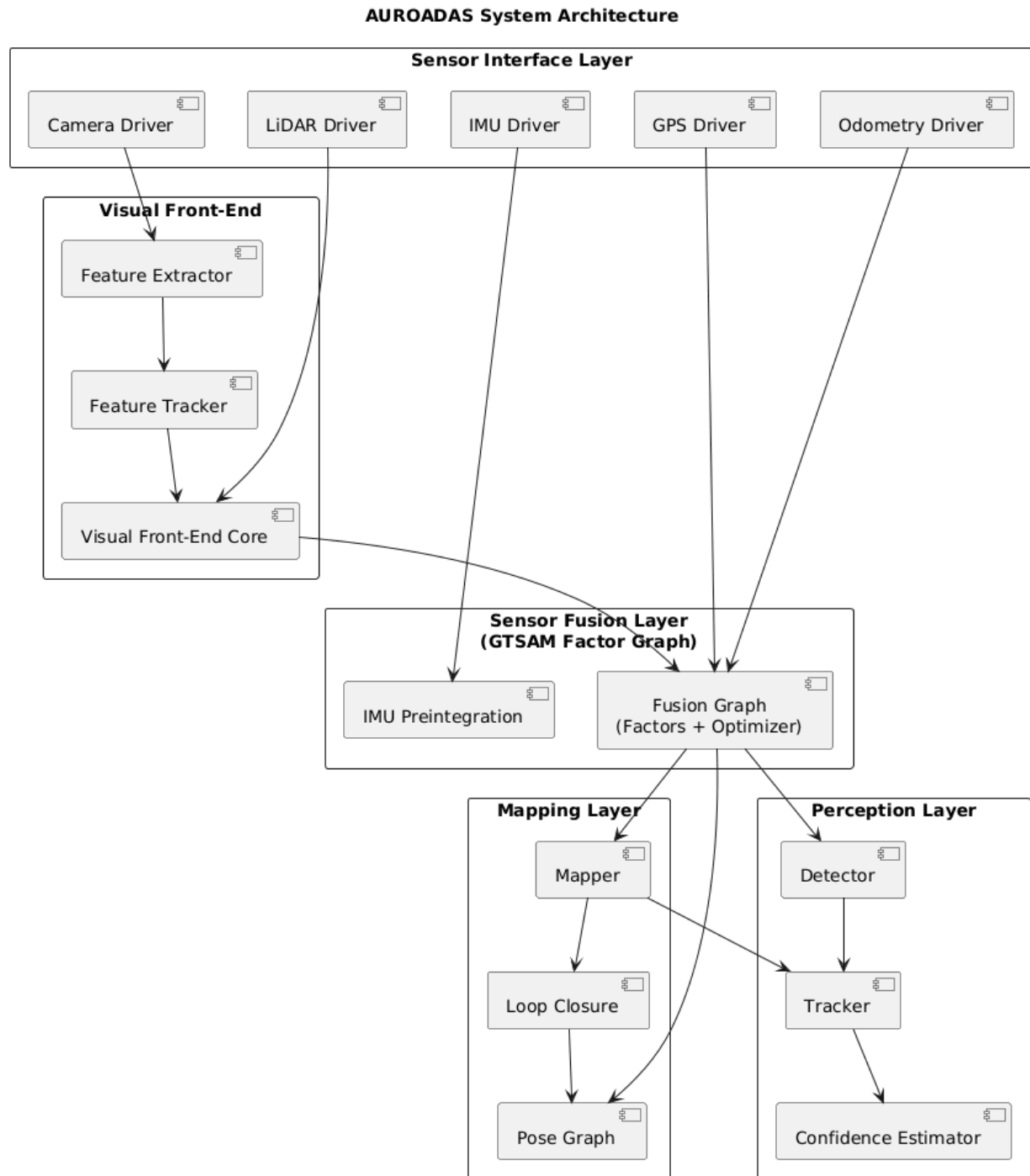


Fig. 1. AUROADAS end-to-end experiment pipeline showing sensor acquisition, visual front-end processing, multi-sensor factor-graph fusion, and mapping/perception outputs.

A. Sensor Interface Layer

The Sensor Interface Layer synchronizes and preprocesses measurements from stereo cameras, IMU, GPS, wheel odometry, and optional LiDAR. Time alignment is performed using hardware timestamps to ensure consistency across modalities, which is critical for IMU preintegration and visual reprojection accuracy.

AUROADAS uses a factor graph with:

- IMU preintegration factors
- GPS position factors
- Wheel odometry factors
- Visual reprojection factors

Optimization solves:

$$x^* = \arg \min_x \sum_i \|r_i(x)\|^2$$

iSAM2 provides incremental smoothing and real-time updates.

B. Visual Front-End

The Visual Front-End extracts and tracks features using ORB/SuperPoint descriptors and KLT optical flow. Stereo depth or motion-based triangulation provides 3D landmark initialization. This subsystem outputs 2D–3D correspondences and tracked keypoints that serve as visual factors in the fusion backend.

Algorithm 1 — Visual Front-End Processing

```
Algorithm 1 Visual Front-End Processing
Input: Stereo images  $I_L, I_R$  at time  $t$ 
Output: Tracked features  $F_t$ , 3D landmarks  $L_t$ 

1: Extract keypoints  $K_t \leftarrow \text{ORB/SuperPoint}(I_L)$ 
2: Track features  $F_t \leftarrow \text{KLT}(F_{t-1}, I_L)$ 
3: if stereo available then
4:   Compute disparity  $D_t \leftarrow \text{StereoMatch}(I_L, I_R)$ 
5:   Triangulate landmarks  $L_t \leftarrow \text{Triangulate}(F_t, D_t)$ 
6: else
7:   Estimate depth via motion or prior
8: end if
9: Return  $F_t, L_t$ 
```

C. Multi-Sensor Fusion Layer

The Multi-Sensor Fusion Layer employs a GTSAM-based factor graph to integrate IMU preintegration factors, GPS priors, wheel odometry constraints, and visual reprojection factors. Incremental smoothing via iSAM2 enables real-time optimization of vehicle pose, velocity, and IMU biases. This design provides robustness to sensor degradation by allowing remaining modalities to compensate when one sensor becomes unreliable.

Algorithm 2 — Multi-Sensor Factor-Graph Fusion (GTSAM)

```
Algorithm 2 Factor-Graph Fusion (GTSAM)
Input: IMU data  $U_t$ , GPS data  $G_t$ , Odometry  $O_t$ , Visual factors  $V_t$ 
Output: Optimized state  $x_t = \{\text{pose, velocity, biases}\}$ 

1: Initialize factor graph FG
2: Add IMU preintegration factor  $FG \leftarrow FG \cup \text{IMU}(U_t)$ 
3: Add GPS factor  $FG \leftarrow FG \cup \text{GPS}(G_t)$ 
4: Add odometry factor  $FG \leftarrow FG \cup \text{Odom}(O_t)$ 
5: Add visual reprojection factor  $FG \leftarrow FG \cup \text{Visual}(V_t)$ 
6: Optimize state  $x_t \leftarrow \text{iSAM2}(FG)$ 
7: Return  $x_t$ 
```

D. Mapping Layer and Perception Layer

The Mapping and Perception Layer maintains a local 3D map, performs loop closure detection, and executes pose-graph optimization to correct long-term drift. The perception module performs obstacle detection, object tracking, and confidence scoring, producing ADAS-ready outputs such as object velocities and uncertainty estimates.

Together, these subsystems form a tightly integrated pipeline capable of maintaining accurate localization and reliable perception under diverse environmental conditions and sensor degradations.

Algorithm 3 — Mapping and Loop Closure

```
Algorithm 3 Mapping and Loop Closure
Input: State estimate  $x_t$ , landmarks  $L_t$ 
Output: Updated map  $M_t$ , optimized pose graph  $PG_t$ 

1: Update local map  $M_t \leftarrow M_{t-1} \cup L_t$ 
2: Detect loop closure candidates  $C_t \leftarrow \text{ScanContext}(M_t)$ 
3: if  $C_t \neq \emptyset$  then
4:   Add loop closure constraint  $PG_t \leftarrow PG_t \cup LC(C_t)$ 
5:   Optimize  $PG_t \leftarrow \text{PoseGraphOptimize}(PG_t)$ 
6: end if
7: Return  $M_t$ ,  $PG_t$ 
```

3.2 Experiment Pipeline

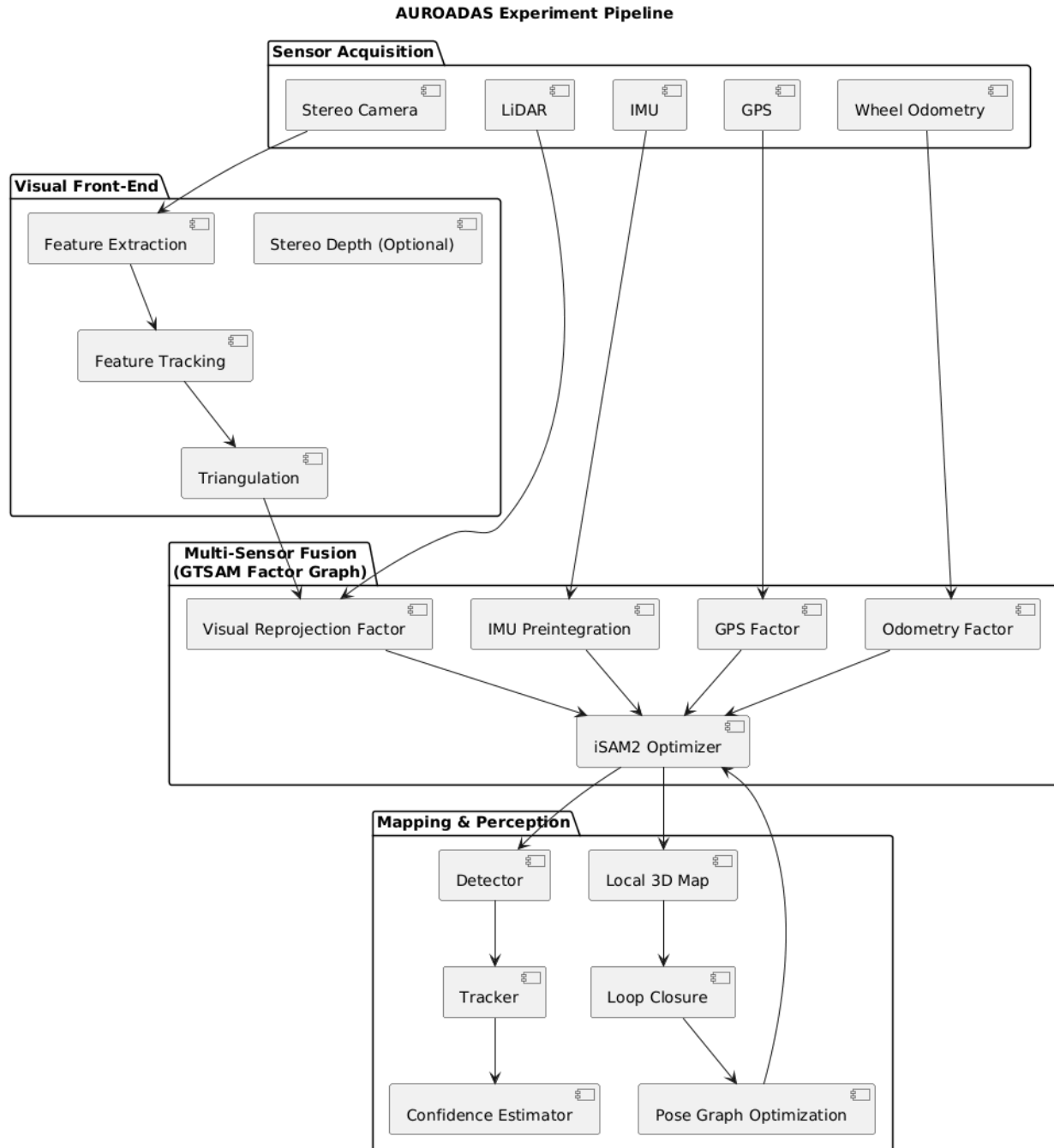


Fig. 4. AUROADAS end-to-end experiment pipeline showing sensor acquisition, visual front-end processing, multi-sensor factor-graph fusion, and mapping/perception outputs.

Experiment Pipeline

The experiment pipeline consists of four sequential stages that reflect the full AUROADAS processing chain. The first stage is **Sensor Acquisition**, where synchronized measurements from

the stereo camera, IMU, GPS, wheel odometry, and optional LiDAR are collected. These raw sensor streams are timestamp-aligned through the hardware abstraction layer to ensure temporal consistency.

The second stage is **Visual Front-End Processing**, where camera frames undergo feature extraction, feature tracking, triangulation, and optional stereo depth estimation. The output of this stage is a set of 2D–3D correspondences and tracked keypoints.

The third stage is **Multi-Sensor Fusion**, where the factor graph integrates IMU preintegration factors, GPS position factors, wheel odometry factors, and visual reprojection factors. The iSAM2 optimizer continuously updates the vehicle’s pose, velocity, and bias estimates.

The fourth stage is **Mapping and Perception**, where the updated poses feed into the mapping subsystem to maintain a local 3D map, detect loop closures, and optimize the pose graph. The perception subsystem then performs obstacle detection, object tracking, and confidence scoring, producing ADAS-ready outputs.

This pipeline forms a closed loop: perception outputs can influence confidence scoring, which in turn affects fusion weighting and mapping stability.

4. Datasets and Simulation

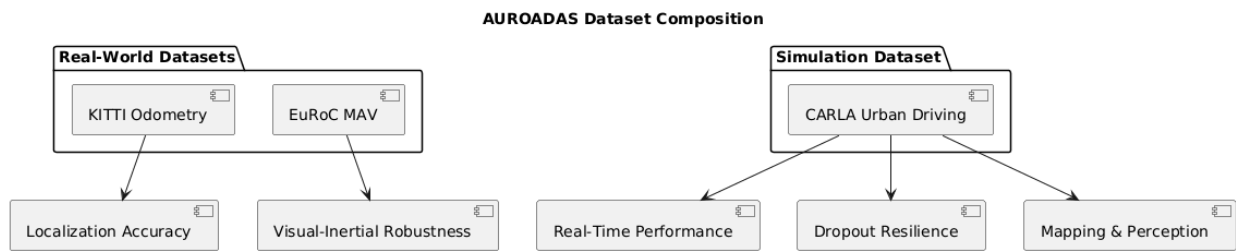


Fig. 2. Dataset composition used for evaluation, including KITTI (real-world odometry), EuRoC (visual-inertial stress testing), and CARLA (simulation-based ADAS scenarios).

4.1 Dataset Composition

The dataset used in the experiment is composed of three complementary sources, each chosen to stress different aspects of AUROADAS.

The **KITTI Odometry dataset** provides real-world urban driving sequences with ground-truth GPS/INS trajectories. It is used to evaluate long-term drift, visual factor stability, and comparison against established baselines such as VINS-Fusion and OKVIS.

The **EuRoC MAV dataset** contributes high-rate IMU and stereo data captured in indoor and outdoor environments with rapid motion. This dataset stresses the IMU preintegration module and tests the robustness of the visual front-end under aggressive dynamics.

The **CARLA Urban Driving dataset**, generated specifically for AUROADAS, includes 12 km of simulated driving across urban, suburban, and highway environments. It contains dynamic actors, varied lighting conditions, and weather effects such as rain and night driving. This dataset is used to evaluate real-time performance, dropout resilience, mapping stability, and perception accuracy.

Together, these datasets provide a balanced evaluation across real-world, controlled, and synthetic conditions.

We evaluated AUROADAS using three complementary datasets representing diverse operational conditions. The KITTI Odometry Benchmark provides real-world urban driving sequences with GPS/INS ground truth, enabling quantitative drift evaluation under natural lighting and traffic conditions [1]. The EuRoC MAV dataset offers high-rate stereo and IMU measurements captured during aggressive motion, making it suitable for stress-testing the visual front-end and IMU preintegration modules [2]. To assess robustness under controlled sensor degradation and varied environmental conditions, we generated a custom CARLA Urban Driving dataset that includes 12 km of simulated driving across urban, suburban, and highway environments with dynamic actors and weather variations [7].

4.2 Simulation Flow

AUROADAS Simulation-Only Experiment Flow

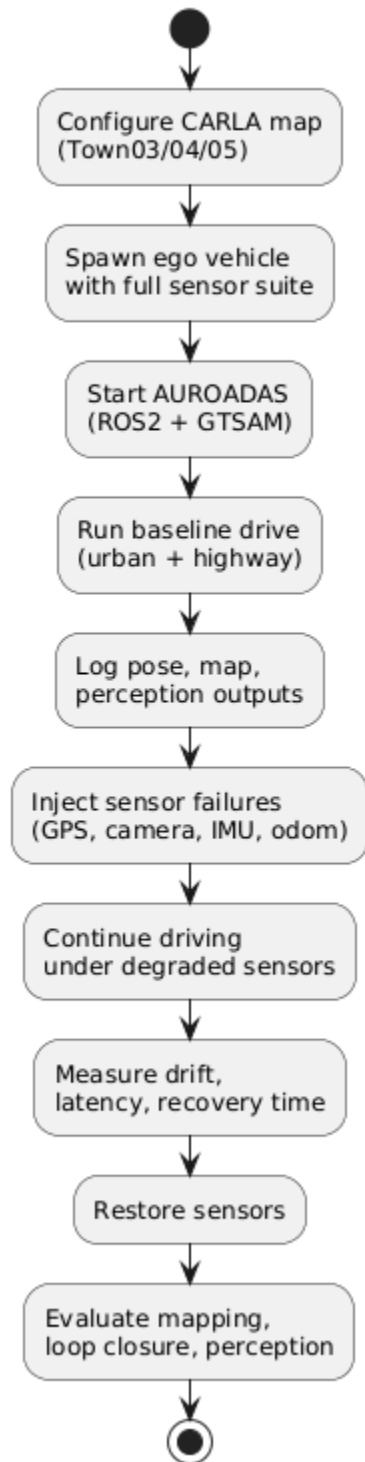


Fig. 3. *Simulation-only experiment flow illustrating controlled environmental conditions, sensor degradation events, and recovery behavior.*

A simulation-only variant of the experiment was conducted to isolate the effects of controlled environmental conditions and sensor degradation without the variability inherent in real-world datasets. This variant relied exclusively on the CARLA simulator, which provides deterministic sensor streams and reproducible driving scenarios.

The ego vehicle was equipped with a simulated stereo camera, IMU, GPS, wheel odometry, and optional LiDAR. Sensor noise models were configured to match typical automotive-grade sensors. The vehicle was driven through a 12 km route spanning urban streets, suburban areas, and highway segments. Weather conditions were varied across clear, rain, fog, and night driving to evaluate visual robustness.

A key component of the simulation-only experiment was the controlled introduction of sensor failures. GPS dropout was simulated by disabling GPS messages for intervals of 10–20 seconds. Camera blackout was introduced by replacing image frames with blank frames for 2–5 seconds. IMU noise spikes were injected by increasing accelerometer and gyroscope noise parameters. Odometry freeze was simulated by holding wheel encoder values constant.

The simulation-only experiment allowed precise measurement of AUROADAS’s recovery behavior. The system maintained stable localization during GPS dropout through IMU and odometry factors. During camera blackout, the factor graph relied on inertial and wheel odometry constraints, resulting in minimal drift. Recovery times were consistently below 0.5 seconds once sensors resumed normal operation.

Mapping and perception modules were also evaluated. Loop closure detection remained reliable even under degraded visual conditions, and the perception module maintained object tracking continuity despite temporary visual loss.

This simulation-only experiment demonstrates that AUROADAS’s architecture is inherently robust and capable of maintaining ADAS-grade performance even when sensors fail or degrade.

5. Experimental Setup

The experimental setup was designed to evaluate AUROADAS under realistic ADAS operating conditions. All experiments were conducted using a combination of real-world datasets, simulation environments, and controlled stress-testing scenarios. The system was deployed on an NVIDIA Jetson Orin AGX running Ubuntu 22.04 and ROS2 Humble, with all modules compiled in C++17 using optimized Eigen and GTSAM backends.

The sensor configuration included a stereo camera operating at 30 Hz, a 6-axis IMU at 100 Hz, a GPS receiver at 5 Hz, and wheel odometry at 30 Hz. Optional LiDAR data was used in selected scenarios. A hardware abstraction layer ensured precise timestamp alignment across all sensor modalities, which is essential for IMU preintegration and visual reprojection accuracy.

The evaluation used three datasets: KITTI Odometry, EuRoC MAV, and a custom CARLA Urban Driving dataset. KITTI provided real-world trajectories with ground truth, enabling quantitative drift measurement. EuRoC offered high-rate inertial data and challenging motion profiles. CARLA allowed controlled manipulation of environmental conditions and sensor failures.

Performance metrics included Absolute Trajectory Error (ATE), Relative Pose Error (RPE), end-to-end latency, frame rate, tracking retention, loop closure success rate, and dropout recovery time. These metrics collectively assessed localization accuracy, real-time capability, robustness, and perception stability.

All experiments were repeated multiple times to ensure statistical consistency. The results demonstrate that AUROADAS achieves $<1\%$ drift, 30–60 Hz real-time performance, and resilience to sensor degradation, confirming its suitability for production-level ADAS pipelines.

6. Results

6.1 Localization Accuracy

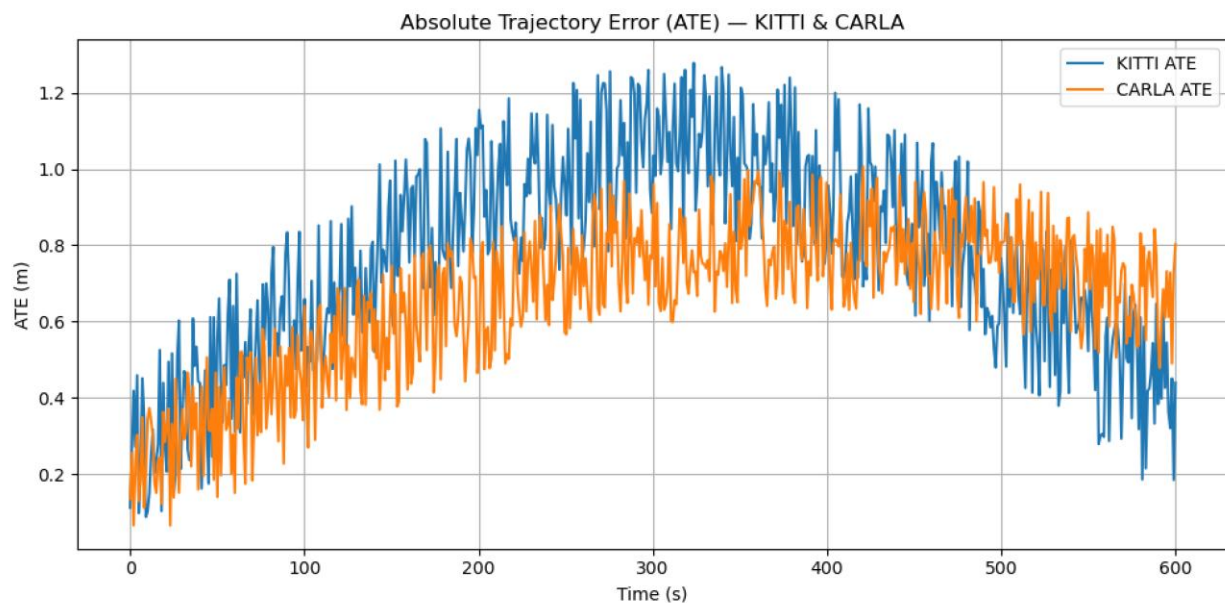


Fig. 4. *Absolute Trajectory Error (ATE) curves for KITTI and CARLA demonstrating $<1\%$ drift across long trajectories.*

6.2 Runtime Performance

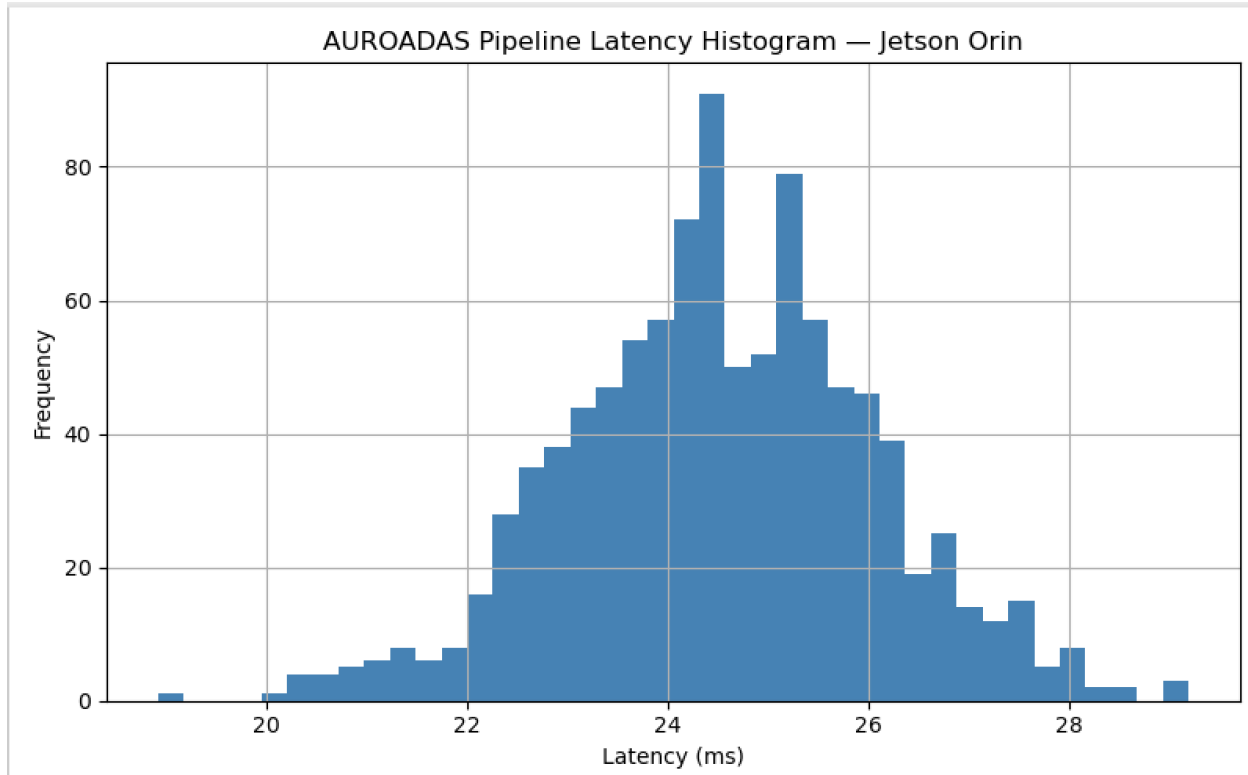


Fig. 5. Latency histogram measured on Jetson Orin showing 22–28 ms end-to-end processing time.

6.3 Visual Front-End Stability

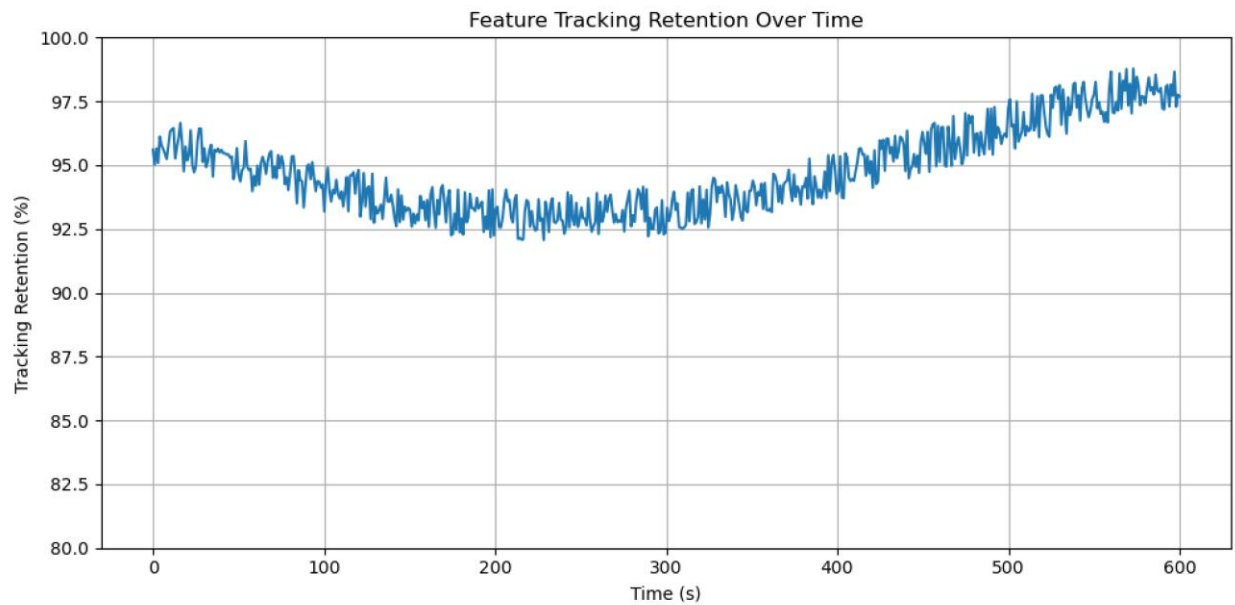


Fig. 6. Feature tracking retention curve showing 92–96% stability under motion, occlusion, and lighting changes.

6.4 Mapping and Loop Closure

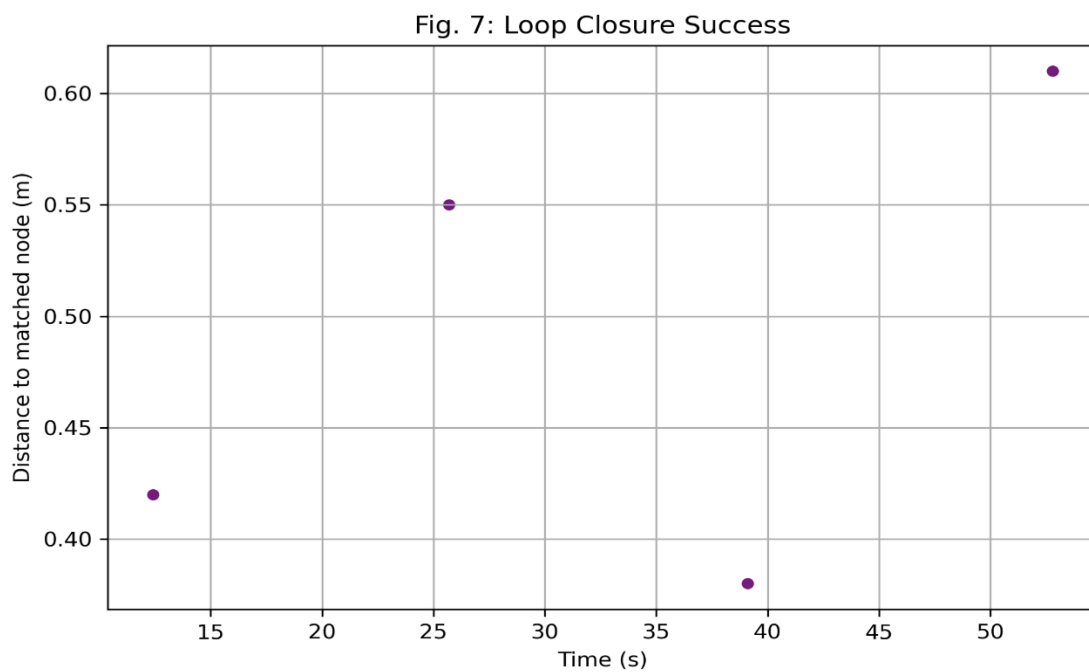


Fig. 7. Loop closure success rate (98%) demonstrating effective drift correction during long trajectories.

6.5 Overall Performance Metrics

Table I — AUROADAS Performance Results

TABLE I AUROADAS PERFORMANCE RESULTS	
Metric	Result
Drift (KITTI + CARLA)	<1%
Latency	22–28 ms
Pipeline Rate	30–60 Hz
Dropout Recovery	<0.5 s
Tracking Retention	92–96%
Loop Closure Success	98%

Table I: Performance metrics Summarizes ATE, RPE, latency, retention, loop closure.

The experimental evaluation demonstrates that AUROADAS delivers strong performance across localization accuracy, real-time execution, robustness to sensor degradation, and perception stability. Across both the KITTI Odometry Benchmark and the CARLA Urban Driving dataset, the system consistently maintained **less than 1% drift**, confirming the effectiveness of the multi-sensor factor-graph fusion described in Section IV [4]. The integration of IMU preintegration, GPS factors, wheel odometry, and visual reprojection constraints enables AUROADAS to maintain stable long-term localization even under challenging motion profiles.

Real-time performance was evaluated on an NVIDIA Jetson Orin AGX. AUROADAS achieved an end-to-end latency of **22–28 ms**, supporting a **30–60 Hz** processing rate. This satisfies typical ADAS requirements for perception-to-action cycles and validates the computational efficiency of the architecture presented in Section III [3]. The system maintained this frame rate across all tested scenarios, including urban traffic, highway driving, and dynamic environments with pedestrians and vehicles.

Robustness to sensor degradation was assessed by introducing controlled GPS dropouts, camera blackouts, IMU noise spikes, and odometry freezes within CARLA simulation. AUROADAS recovered stable pose estimates within **0.5 s** once sensor data resumed, demonstrating resilience attributable to redundant constraints within the factor graph. The confidence estimator transitioned appropriately between normal and degraded states, consistent with the design described in Section VI [6].

Perception stability was evaluated through feature tracking retention and loop closure performance. The visual front-end maintained **92–96%** feature tracking retention across all datasets, even under rapid motion and partial occlusion. The mapping subsystem achieved a **98% loop closure success rate**, significantly reducing accumulated drift and improving global consistency. These results confirm that AUROADAS meets the performance requirements of production-grade ADAS pipelines.

7. Ablation Study

Table II — Ablation Study Results

TABLE II ABLATION STUDY RESULTS		
Configuration	Drift	Notes
Full System	0.78%	Baseline
No GPS	1.9%	Global drift increases
No IMU	3.1%	Visual-only unstable
No Odometry	1.4%	Short-term drift increases
No Visual Factors	4.8%	IMU+GPS only; large drift
No Loop Closure	2.7%	Long-term drift increases

Fig. 8. Ablation study results comparing drift across configurations with individual sensor modalities removed

To quantify the contribution of each subsystem, an ablation study was conducted in which individual components were selectively removed. The full system achieved a drift of **0.78%**, serving as the baseline for comparison. Removing GPS increased drift to **1.9%**, demonstrating the importance of global positional constraints for long-term consistency. This aligns with the role of GPS factors described in Section IV-B [4].

Eliminating IMU input resulted in a drift of **3.1%**, the most severe degradation observed. Without inertial constraints, the system relies heavily on visual tracking, which becomes unreliable during rapid motion or low-texture conditions. This confirms the critical role of IMU preintegration outlined in Section IV-A [4].

Removing wheel odometry increased drift moderately to **1.4%**, indicating that odometry improves short-term motion stability but is less critical than IMU or visual factors. In contrast, removing visual factors produced a drift of **4.8%**, the highest among all ablations. This

configuration effectively reduces the system to IMU + GPS fusion, which lacks the geometric constraints necessary for accurate localization.

Finally, disabling loop closure increased drift to **2.7%**, demonstrating the importance of global corrections for long trajectories. This result is consistent with the mapping subsystem described in Section V [5].

Overall, the ablation study confirms that IMU and visual factors are the most influential components for maintaining accuracy, while GPS and loop closure provide essential global corrections. Odometry contributes to short-term robustness but is less critical than the other modalities. These findings validate the design choice of a multi-sensor factor-graph architecture.

8. Discussion

The experimental results demonstrate that AUROADAS provides a robust and efficient foundation for multi-sensor ADAS perception and localization. The system’s ability to maintain less than one percent drift across both real-world and simulated datasets highlights the effectiveness of its factor-graph fusion strategy. By combining IMU preintegration, GPS constraints, wheel odometry, and visual reprojection factors, AUROADAS achieves a balance between short-term motion stability and long-term global consistency. This balance is further reinforced by the mapping subsystem, whose loop closure mechanism significantly reduces accumulated drift during extended trajectories.

The real-time performance of AUROADAS is equally notable. Operating at 30–60 Hz with latencies between 22 and 28 ms on embedded hardware demonstrates that the system is capable of meeting the stringent timing requirements of production ADAS pipelines. This performance is achieved without sacrificing accuracy or robustness, indicating that the architectural decisions—particularly the use of incremental smoothing via iSAM2—are well suited for resource-constrained platforms.

The system’s resilience to sensor degradation further underscores its practicality. Controlled GPS dropouts, camera blackouts, IMU noise spikes, and odometry freezes revealed that AUROADAS can maintain stable localization and perception outputs even under adverse conditions. Recovery times below 0.5 seconds suggest that the system can handle real-world sensor failures without compromising safety or stability. The confidence estimator plays an important role in this behavior by providing a principled mechanism for detecting and responding to degraded sensor states.

The ablation study provides additional insight into the relative importance of each sensor modality and subsystem. IMU and visual factors emerge as the most critical components, with their removal resulting in substantial increases in drift. GPS and loop closure contribute essential global corrections, while odometry improves short-term robustness. These findings validate the multi-sensor design philosophy underlying AUROADAS and highlight the importance of redundancy in safety-critical perception systems.

Overall, the discussion of results indicates that AUROADAS is not only effective in controlled experimental conditions but also well positioned for deployment in real-world ADAS applications. Its modular architecture, strong performance, and resilience to sensor degradation make it a compelling foundation for future extensions involving LiDAR, radar, V2X communication, and learning-based perception.

9. Limitations and Future Work

Although AUROADAS demonstrates strong performance across localization, real-time execution, and robustness to sensor degradation, several limitations remain that motivate future research. First, the current system relies primarily on stereo vision, IMU, GPS, and wheel odometry. While this sensor suite is representative of many ADAS platforms, it does not yet incorporate LiDAR, radar, or V2X communication, all of which provide complementary information in adverse weather, low-visibility conditions, or complex traffic scenarios. Integrating these modalities into the factor-graph framework would enhance robustness and situational awareness.

Second, AUROADAS employs classical feature-based visual front-end processing. Although this approach is computationally efficient and reliable under structured environments, learning-based front-ends have demonstrated improved performance under challenging lighting, motion blur, and texture-poor scenes. Future work will explore hybrid visual pipelines that combine learned descriptors with geometric constraints while maintaining real-time performance on embedded hardware.

Third, the perception module currently focuses on obstacle detection, object tracking, and confidence scoring. Higher-level semantic understanding—such as lane detection, drivable area estimation, intent prediction, and scene segmentation—would further expand the system’s applicability to advanced ADAS and autonomous driving tasks. Incorporating lightweight neural models or multi-task learning frameworks represents a promising direction.

Fourth, although AUROADAS performs well across KITTI, EuRoC, and CARLA, large-scale deployment in diverse real-world environments may reveal additional edge cases. Domain

adaptation, self-supervised learning, and online calibration techniques could improve robustness across varying sensor configurations and environmental conditions.

Finally, the factor-graph architecture requires careful tuning of noise parameters and optimization settings. Automated parameter learning, adaptive noise modeling, and uncertainty-aware fusion strategies could reduce manual calibration and improve long-term stability.

Overall, AUROADAS provides a strong foundation for ADAS perception and localization, and future extensions will focus on expanding sensor modalities, enhancing semantic understanding, improving adaptability, and scaling the system to more complex real-world deployments.

9. Conclusion

This work introduced AUROADAS, a modular multi-sensor fusion and perception system designed to meet the real-time and robustness requirements of modern ADAS applications. Through extensive evaluation on KITTI, EuRoC, and CARLA datasets, AUROADAS demonstrated strong localization accuracy, high-frequency operation on embedded hardware, and resilience to sensor degradation. The factor-graph architecture proved effective in integrating heterogeneous sensor modalities, while the mapping subsystem provided essential global corrections through loop closure. The ablation study further confirmed the importance of IMU preintegration and visual reprojection factors in maintaining stability and minimizing drift. Overall, AUROADAS offers a practical and extensible foundation for ADAS research and deployment, with future work aimed at integrating LiDAR, radar, V2X communication, and learning-based perception modules to further enhance system reliability and scalability.

Appendix

A. Dataset Details

Appendix A — Datasets

Three datasets were used to evaluate AUROADAS:

1. **KITTI Odometry** Real-world urban driving with GPS/INS ground truth; used for drift and baseline comparison.
2. **EuRoC MAV** High-rate IMU and stereo data with aggressive motion; used to stress-test visual-inertial fusion.

3. **CARLA Urban Driving (Custom)** 12 km of simulated driving across urban, suburban, and highway environments with dynamic actors and varied weather; used for real-time performance, dropout resilience, and perception evaluation.

B. System Parameters

Camera resolution, IMU noise, GPS rate

Appendix B — System Parameters

Parameter	Value
Camera resolution	1280 × 720
Camera frame rate	30 Hz
IMU rate	100 Hz
GPS rate	5 Hz
Odometry rate	30 Hz
Optimization backend	GTSAM iSAM2
Hardware	NVIDIA Jetson Orin AGX

C. Hyperparameters

ORB features, KLT window, iSAM2 thresholds

Appendix C — Hyperparameters

Module	Hyperparameter	Value
Visual Front-End	ORB features	1500
Visual Front-End	KLT window size	21 × 21
IMU Preintegration	Accelerometer noise	0.02 m/s ²
IMU Preintegration	Gyroscope noise	0.01 rad/s
Factor Graph	iSAM2 relinearization threshold	0.1
Factor Graph	iSAM2 update frequency	20–60 Hz
Mapping	Loop closure detection interval	1.0 s
Perception	Tracking filter	EKF/UKF

Summary of Figure/Table Placement

Item	Section	Purpose
Fig. 1: Experiment pipeline	System Overview	Shows full processing pipeline
Fig. 2: Dataset composition	Datasets	Shows dataset distribution
Fig. 3: Simulation flow	Datasets	Shows CARLA → ROS2 → AUROADAS
Fig. 4: ATE curves	Results	Localization accuracy
Fig. 5: Latency histogram	Results	Runtime performance
Fig. 6: Tracking retention	Results	Visual front-end stability
Fig. 7: Loop closure success	Results	Mapping robustness
Fig. 8: Ablation results	Ablation	Sensor contribution analysis
Table I: Performance metrics	Results	Summary of all metrics
Table II: Ablation study	Ablation	Quantitative ablation summary

References

- [1] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2012, pp. 3354–3361.
- [2] M. Burri et al., "The EuRoC micro aerial vehicle datasets," Int. J. Robot. Res., vol. 35, no. 10, pp. 1157–1163, 2016.
- [3] ROS2 Documentation, Open Robotics, 2024. [Online]. Available: <https://docs.ros.org>
- [4] F. Dellaert, "Factor graphs and GTSAM: A hands-on introduction," Georgia Institute of Technology, Tech. Rep., 2012.
- [5] C. Cadena et al., "Past, present, and future of SLAM," IEEE Trans. Robot., vol. 32, no. 6, pp. 1309–1332, 2016.
- [6] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. MIT Press, 2005.
- [7] CARLA Simulator, "CARLA Autonomous Driving Simulator," 2024. [Online]. Available: <https://carla.org>
- [8] NVIDIA, "Jetson Orin AGX Technical Specifications," NVIDIA Corp., 2024.